TECHNICAL REPORT | COMMENTS INCORPORATED | VERSION 3

PV asset: the world, the agent and experience

Technical report v3 Who changes the world, who decides, and how experience could guide future actions

Version 3 - 19 October 2026

> **Evidence status**
> The WORLD is currently a simulator outside the decision-making AGENT. Current agent procedures and records are implemented; richer internal beliefs and retrieval of past experience are proposed. Real hardware transfer remains unvalidated.

The distinction: the world is the PV plant and its surroundings; the agent is the software that observes and chooses actions. The simulator's equations belong to the world implementation. An agent world model, if added, belongs inside the agent and expresses its beliefs about that external world.

Your memory proposal: yes, past observations, choices and outcomes could guide a later investigation through retrieval and reasoning without training new model weights. Recording already exists. Automatic retrieval, applicability checking and experience-based improvement have not yet been implemented or demonstrated.

This revision addresses all four comments in the annotated v2: clearer world/agent boundaries, a concrete memory pathway, and role-labelled source-change and B1 action traces. The annotated v2 is preserved; Section 12 maps each comment to its revision.

Evidence remains unchanged: the source-change fixture, B1 pilot and authored information fixture are separate demonstrations. The 54-case focused test result is retained from v2, not a fresh v3 test run. New figures replot existing records; no new agent, simulation, rendering-engine or hardware experiment is introduced.

Contents

Sections 1-4 define the world, interfaces, agent tools and internal memory/models. Sections 5-7 show role-labelled evidence. Sections 8-11 retain the testing and development programme; Section 12 answers each v2 comment.

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1. Start with four roles: world, interface, agent and evaluator

The WORLD is the PV panel, camera/load hardware and surrounding illumination or thermal conditions. The AGENT is decision-making software acting for the PV asset. The boundary is functional: even if a processor is eventually embedded inside a module, the physical plant evolves while the agent observes, reasons and issues requests. The agent does not directly choose the measured outcome.

Table 1. Vocabulary used consistently throughout v3.

Role	Meaning	Current implementation
WORLD / plant	The external system being observed and acted on.	MuJoCo camera dynamics and renderer plus optical, PV and thermal equations. These non-agent equations implement the simulated environment.
INTERFACE	The sensors and permitted action routes between agent and world.	Timestamped public measurements, typed actions, capability declarations, limits and measured feedback.
AGENT	The component that receives evidence and selects a next action.	B0/B1 local policies. A separate authored Step 2 driver exercises information requests; it is not an adaptive reasoning agent.
EVALUATOR	The experimental observer that checks what really happened and scores the agent.	Private scenario/state logs, independent reference calculations and tests. Private truth is excluded from policy input.

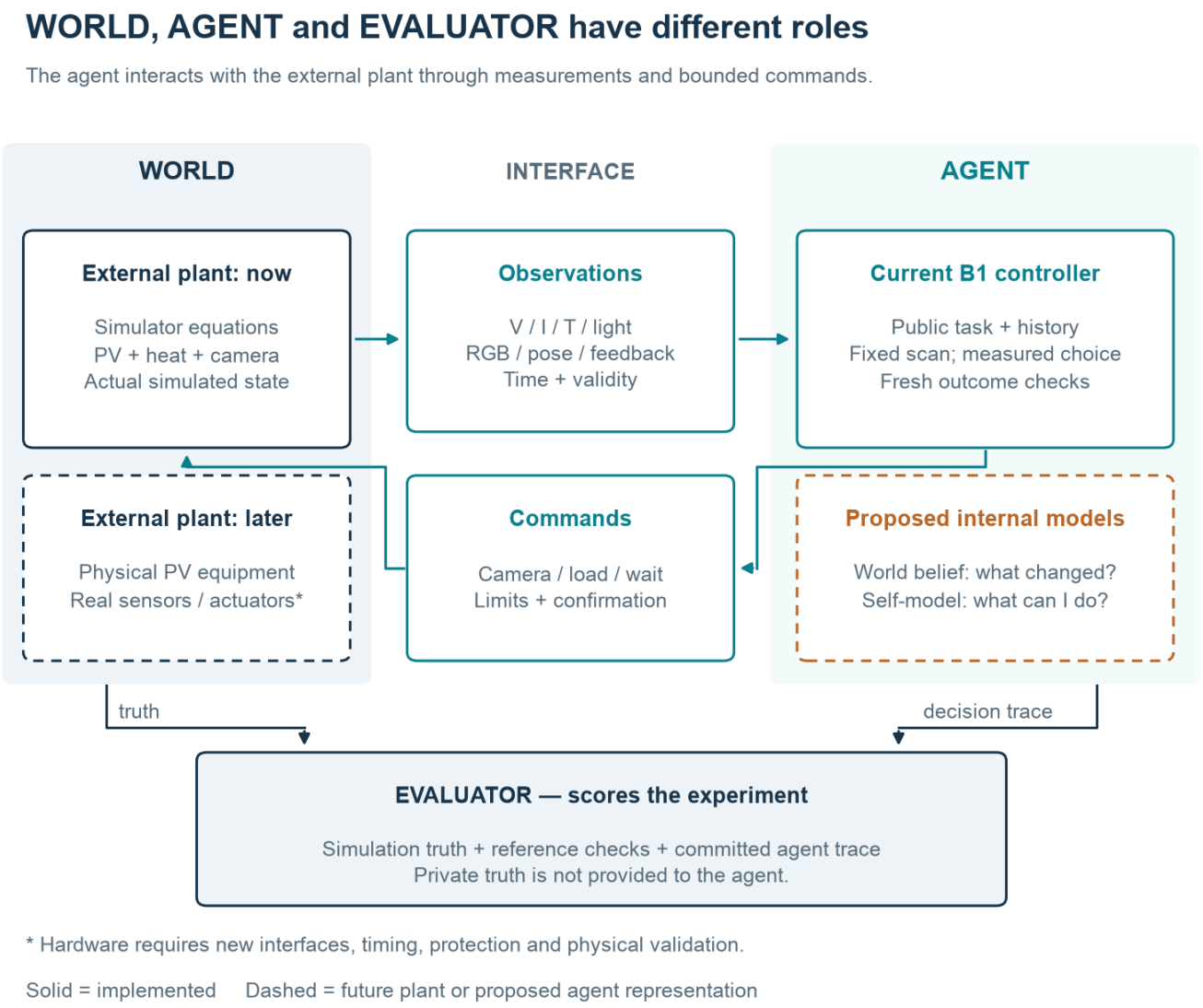


Figure 1. The WORLD and AGENT are distinct components connected by observations and bounded commands. The evaluator is a separate assessment role. Dashed internal agent capabilities and real-hardware substitution are proposed; they are not current runtime features.

Current contracts and capabilities [A1, A4-A6]; v3 explanatory diagram [A8].

Replacing simulation with reality

Conceptually, the external simulated plant can be replaced by a real panel, sensors and actuators. The decision role and abstract observe/act loop can remain. The hardware adapter, calibration, protection, units and timing must be implemented and validated; this is not a completed plug-and-play substitution. A real panel cannot pause while the agent waits for a human.

Table 2. Read the experiment as a sequence of claims.

Stage	Question to answer	Evidence required
World changes	Did an actual model variable change?	Private event/state trace and independent physical calculation.

Stage	Question to answer	Evidence required
Sense	What could the agent observe, and when?	Public values/images, validity, capture and delivery times.
Analyse	What conclusion did the agent derive?	Agent-produced interpretation and cited observations; competing explanations where needed.
Choose and act	Why this interaction or operating action?	Prior decision, expected effect, permitted action and actual application.
Verify	Did the expected response occur?	New independent-of-command sensor evidence, outcome and uncertainty.
Evaluate	Was the decision correct and useful?	Evaluator truth revealed only after the decision; costs and failure cases.

In simulation, the evaluator can inspect complete private state. In a physical experiment, it would instead use independent reference measurements and controlled interventions, with uncertainty; full physical truth is not generally available. Neither form of privileged evaluation evidence should silently enter the agent's input or memory.

The current source-change fixture establishes WORLD response and SENSOR evidence. The separate B1 episode establishes a narrow AGENT observe/choose/act/check loop. The information fixture establishes request/response mechanics. No accepted single episode yet links a hidden event to adaptive interpretation, choice of information source and verified outcome. [A1-A3]

2. WORLD: the external plant and its hidden changes

This section describes the non-agent environment simulator. It implements the external WORLD: physical state evolves on a 2 ms clock, with equations for PV, thermal and optical effects and finite camera dynamics. The agent supplies permitted voltage/camera requests through the interface. Evaluator-set source events and faults are private inputs; sensors expose selected consequences. These environment equations are not an internal cognitive model available to B1. [A4, R7, R8, R18]

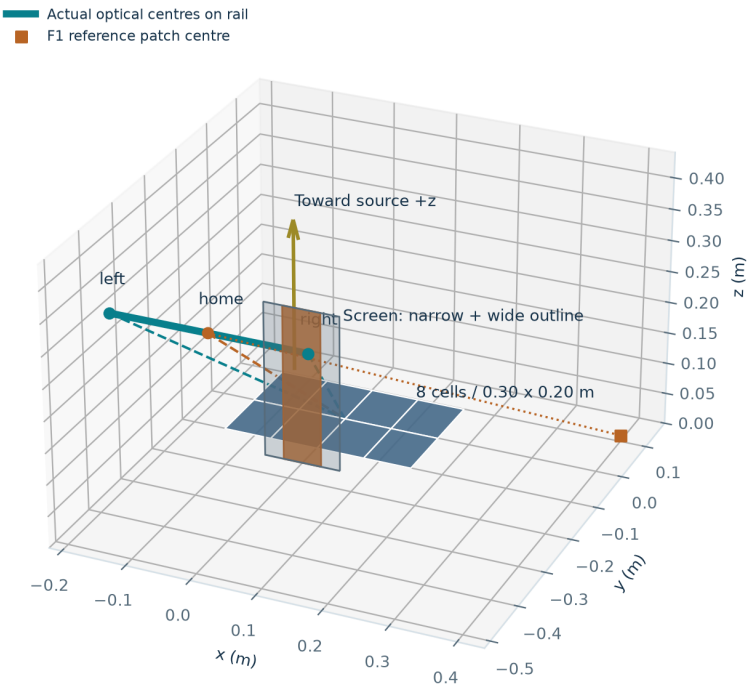


Figure 2. Geometry explanation reused from the accepted M2 report. The rail moves the camera; the panel remains fixed. The narrow and wide screen outlines and the distant F1 reference position combine fixture alternatives for explanation. This is a scientific geometry plot, not an agent camera frame or one simultaneous scene.

Accepted M2 report geometry figure and provenance [R9].

Table 3. Environment mechanisms versus agent access.

Mechanism	Hidden state or intervention	Agent-visible consequence
Illumination	DNI/DHI, direction, per-cell transmission and optional private source-event schedule.	Reference irradiance; changed electrical response; qualitative RGB brightness when represented by the renderer.
Electrical plant	Eight cells in two four-cell substrings, bypass approximation, temperature-sensitive piecewise I-V relation and resistance.	Measured terminal V and I after actual load delay/slew; power can be calculated from matched readings.
Thermal plant	Lumped temperature driven by absorbed light, electrical extraction and ambient heat exchange.	Temperature samples; slower response than instantaneous electrical change.
Camera embodiment	Finite rail/pan/tilt dynamics, pose, reachability and view obstruction.	Encoder measurements and captured RGB. Moving the camera changes viewpoint.

Mechanism	Hidden state or intervention	Agent-visible consequence
Transport and faults	Configured noise, latency/loss/staleness and command inhibition/drop/delay.	Timestamped validity and measured action feedback; missing or failed evidence remains explicit.

The panel is fixed. Current policy actions do not rotate, clean or reconfigure it. M2 view screens block the camera view while leaving illumination unchanged; seeing a screen is not evidence of electrically effective shading. Per-cell transmission is an optical/electrical model input, but it is not automatically rendered as visible dirt. The accepted source step changes illumination and rendered light together under a declared qualitative mapping. [R8, R18]

Private state is not a hidden answer in the public task

PublicTask declares the goal, sensor/action availability, nominal camera calibration, limits, budgets and hypothesis vocabulary. It does not disclose the event schedule, private seed, true per-cell condition or evaluator metrics. Its present knowledge condition is 'uninformed; public nominal calibration and action limits only', and allowed_adaptation is 'none'. The existing schema can hold predictions and hypothesis probabilities, but the baselines do not produce them. [A4-A5]

> **Hidden does not mean unobservable**
> The DNI schedule and event trigger are private; their effects are measurable. The existing reference irradiance sensor legitimately reveals an illumination change. Hiding that available reading just to force an internet or human request would make the comparison misleading.

When moving to hardware, these roles remain distinct: sunlight and the real panel determine the response, actual sensors report it, a controller applies permitted commands, and the agent evaluates the returned evidence. The simulator is a test implementation of the outside world; it need not be embedded inside a future decision agent.

3. AGENT: tools, procedural skills, decisions and memory

The experimental AGENT is a public-observation decision process, not the coding assistant that built the project. Its inputs are the declared task and permitted observation history; its output is a typed Decision. B0 and B1 implement this role today. No foundation-model provider or learned policy is connected in the accepted runs, and the Step 2 authored driver is a separate interface demonstration. [A1, A4-A5, R21]

Table 4. The current controllers and their actual decision rules.

Component	What it does	What it does not establish
B0 FixedPolicy	Hold 1.8 V, confirm it, collect ordinary measurements and finish within the time allowance.	No anomaly detection or diagnosis; no comparison proving it inferior to B1.
B1 ScanPolicy	Capture home view; move/capture side view; probe 0.8, 1.8 and 3.0 V; choose maximum measured V x I; apply and freshly verify.	Sequence is fixed. Voltage selection depends on measurements, but it does not select an investigative channel or diagnose a cause.
Step 2 fixture driver	Issue three authored human/internet requests, consume simulated replies, run a two-point voltage probe and finish unresolved.	Questions, follow-up, interpretation and actions are authored. This tests plumbing, not response-dependent reasoning.

Table 5. Tool, skill, action, memory and model mean different things.

Term	Meaning in the agent	Concrete example
Tool	An available interface for obtaining information or requesting an effect.	RGB capture, encoder/scalar access or the bounded voltage-probe interface.
Skill / procedure	A reusable way to combine tools and check their results.	Pair V/I by capture time; compare measured powers; require fresh verification. These are programmed procedures, not learned skills.
Action	One specific request issued now with a target and ID.	Probe 0.8/1.8/3.0 V, then request 1.8 V and wait for new measurements.
Memory	Evidence retained from an investigation.	Public observations, earlier choices and measured outcomes; persistence exists, automatic reuse is proposed.
Agent model	A representation used to interpret evidence or predict consequences.	Proposed world beliefs and capability beliefs. Their existence should be tested, not inferred from record storage.

Capabilities, tools and skills in concrete terms

Table 6. A capability inventory without attributing unimplemented intelligence.

Layer	Implemented representation	Current use
Sensing tools	V/I/T/reference samples, encoders, RGB when enabled; capture/delivery/validity metadata.	B1 pairs V/I, checks frames exist and checks measured pose. It does not interpret image content.
Physical tools	move_camera, capture_rgb, set_voltage, probe_voltage, wait, finish and safe_stop under declared limits.	Typed commands enter the existing action gate. Probe restores the prior load before completion.
Information tools	Separate interaction/1.0 requests, replies, provenance, follow-ups and information_refs.	Recorded/simulated adapters only. Channel names human/internet do not mean live connectivity.
Reusable procedures	Matched-sample pairing, fixed scan/probe/verify sequence, completion checks, retry and cleanup.	Useful procedural skills in an engineering sense; no learned skill library or skill acquisition.
Memory	Public observation history, pending action, next ID, sequence stage, selected voltage and evidence references.	Within-episode bookkeeping. No persistent learned world knowledge or cross-episode improvement.
Decision boundary	PolicyRunner serializes/revalidates public task/history and validates the returned Decision.	Accidental-leakage prevention for trusted local code. It is not an operating-system sandbox.

Capability should be read from the task at runtime, not assumed from a class name. For example, RGB is absent when rendering is explicitly disabled; the rail can be locked by advertised limits. The legacy request_help action remains a separate measurement contract and is not advertised by these accepted environments. It is not the free-text dialogue channel. [A4, R23]

4. Inside the AGENT: beliefs and experience without new training

The simulator and an agent world model serve different purposes. The simulator generates the environment's actual digital response. An agent world model would represent what the agent currently believes about that response, using only permitted evidence; it may be incomplete or wrong. In a hardware trial, the physical WORLD replaces the simulator while the agent may still maintain its own internal model.

Table 7. Where each representation belongs.

Representation	Owner and function	Status
Environment simulator	WORLD implementation: equations and state produce consequences from disturbances and actions.	Implemented. Not part of B1's reasoning; could later be replaced by measured physical hardware through a validated adapter.
Agent world model	Inside AGENT: beliefs about the panel/environment, competing explanations and expected action effects.	No explicit revisable causal model in B0/B1. A small structured belief record is proposed.
Agent self model	Inside AGENT: knowledge of available sensors/actions, limits/resources and confidence in their effective operation.	Declared capabilities and action bookkeeping exist. Learned or inferred reliability and richer replanning remain proposed.
Experience memory	Inside the agent's permitted data boundary: past observation, decision, action and outcome records that could inform later reasoning.	Records and within-episode history exist. Automatic cross-episode retrieval and reuse are not demonstrated.
Reference model / truth	EVALUATOR: independent calculations or reference measurements used to assess claims.	Digital references exist; physical calibration/validation is separate. Not an agent knowledge source by default.

Here the self model concerns the agent's sensing/actuation access and effectiveness. The world model concerns the physical asset and its surroundings. A failed load actuator can inform both views, but they answer different questions: 'What is happening to the panel?' and 'Can my requested intervention be trusted to occur?' They need not be separate neural networks.

Yes: records can guide future actions without training new weights

The simplest proposed approach is to keep a compact experience record, retrieve a relevant earlier case, compare it with the current situation, and use it to propose a next step. The decision process could be rules or a separately declared pretrained reasoning model whose weights stay fixed. Updating the record or a belief state is different from training neural-network weights. No new model training is required for this first memory mechanism.



Figure 3. Proposed experience-use loop. Recording/replay and within-episode history already exist; retrieval, applicability checks and reuse to guide later decisions are proposed. New sensor evidence closes the loop so that remembered results are tested rather than blindly repeated.

Implemented recording and policy history [A5-A6]; proposed v3 architecture [A8].

Table 8. What a useful experience record would contain.

Field	Purpose and example	Current / proposed
Public context	Sensor values, validity, capture/delivery/return times, pose, advertised tools and budgets.	Recorded. These distinguish situations with different illumination, temperature or capabilities.
Observed gap	What was unexplained, with observation IDs and alternatives.	Information-gap fields exist; an agent-derived anomaly/explanation is proposed.
Requested information	Question, source, response, availability and relation to a decision.	Step 2 records exist with simulated/recorded provenance; semantic interpretation is not demonstrated.

Field	Purpose and example	Current / proposed
Decision and expectation	Chosen tool/action, rationale, evidence references and expected measurable effect/time window.	Decisions recorded. Prediction schema exists, but current B1 predictions are empty.
Applied action and outcome	Measured application/completion, fresh response, failure/restoration and raw costs.	Recorded; distinguish requested targets from measured physical outcomes.
Reusable lesson and applicability	What worked under which conditions, unresolved alternatives and when it should not be reused.	Proposed derived record. It must not copy private evaluator labels or pretend an association proves cause.

For example, a permitted memory could say: 'Under the recorded irradiance and temperature range, this three-point probe delivered the greatest measured power at 1.8 V; fresh V/I later confirmed the requested point.' A later agent might retrieve it as a candidate experiment. It should not conclude '1.8 V is always optimal', or reuse the old reading as current verification. Conditions and actuator performance may have changed.

A stored input-action-output sequence is an experience, not automatically a causal model. A successful action may coincide with changing light; a failed response may mean the command did not apply. The proposed reuse step must check context, timing, provenance and competing explanations, then predict and measure the new outcome. Only public or explicitly permitted information may enter agent memory; evaluator truth, hidden schedules and held-out answers remain separate.

> **What is implemented today**
> Recorder and the information transcript preserve records; PolicyRunner passes public history; B1 uses within-episode measurements. Replay reproduces an existing sequence and RecordedTransport returns supplied exchanges. None of these demonstrates automatic experience search, generalized learning from prior cases or improving future decisions. [A5-A6, R23]

How to test the proposed memory capability

Extend the planned P03/P06/P07 experiments with a no-memory comparison, a relevant prior case, an irrelevant or misleading case, and a held-out related case. Freeze which records are accessible before evaluation. Check whether memory improves a supported decision or reduces investigation cost, whether outdated cases are rejected, and whether predictions are still verified. These tests are proposed; no memory benefit or new training result is claimed here. [A3]

5. WORLD test: hidden illumination change and SENSOR evidence

This is a WORLD/SENSORS experiment, not an agent diagnosis run. The EVALUATOR privately configures a DNI change from 800 to 400 W/m2 at 2.000 s. DHI remains 100 W/m2, so total uniform irradiance changes from 900 to 500 W/m2. The WORLD computes current, power, temperature and rendered light at a fixed 1.8 V load. The scripted fixture issues captures/waits; no B0/B1 or adaptive agent chooses an investigation in this episode. [R18-R20]

Table 9. Who does what in this source-response fixture?

Owner	Operation	Evidence / limit
EVALUATOR	Set the private source schedule and unchanged control.	Knows the intervention; compares responses with independent calculations.
WORLD	Apply changed illumination through optics/electrical/thermal/rendering mechanisms.	Lower current/power; continuous temperature, slower warming; changed light.
SENSORS / interface	Capture values/images, retain their old state and deliver with latency.	Public samples and images are available only at their recorded delivery/return times.
AGENT	No decision policy is run in this fixture.	A future agent could interpret returned evidence and choose a probe or other allowed action. That step is not a result here.

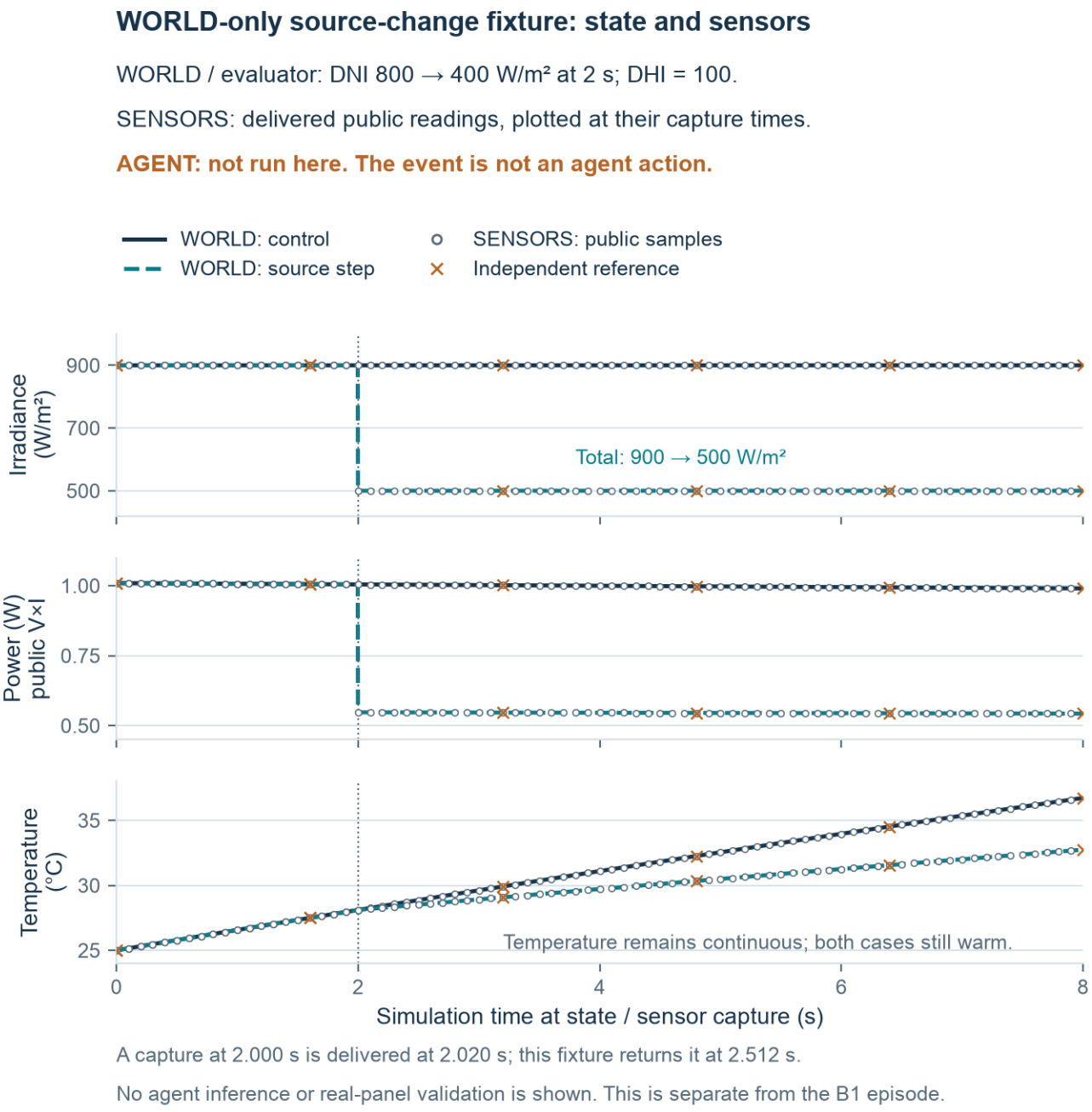


Figure 4. WORLD trajectories after an EVALUATOR-imposed source change, with SENSORS and independent-reference values distinguished. The AGENT is not run here. Measurement markers are positioned at capture time; actual availability remains governed by delivery and observation-return time.

Replot of retained control/treatment truth, public samples and independent reference records [R19, A7].

Table 10. Measured response of the simulated model.

Quantity	Control	Source step
Power at 2.000 s	1.004410089 W	0.547437663 W
Temperature at 2.000 s	28.110258192 °C	28.110258192 °C
Temperature at 8.000 s	36.730084298 °C	32.744402518 °C

Quantity	Control	Source step
Electrical energy, 0-8 s	7.996912570 J	5.284456782 J

Retained evaluator states and energy metrics [R19].

Maximum temperature discrepancy against the independent reference is below 5.92×10^{-7} K; electrical-energy discrepancy is below 1.93×10^{-5} J. These are numerical/model-consistency errors for this fixture, not uncertainty bounds on a real PV panel. The post-step temperature does not drop instantaneously: it continues rising with less absorbed input. [R20]

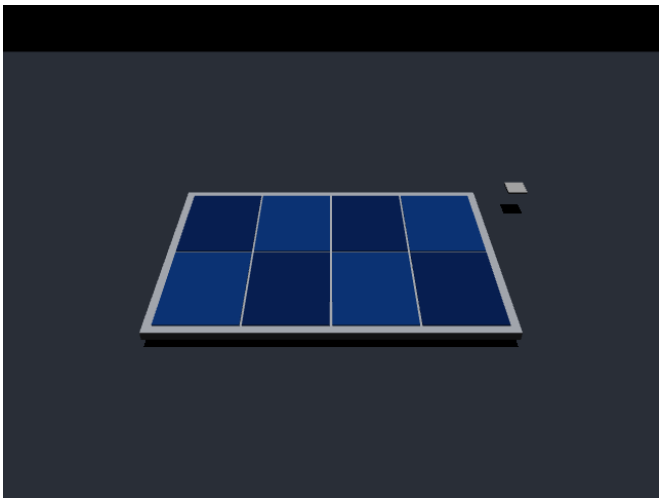
Read the private and public timelines separately

Table 11. Actual public samples around the private event.

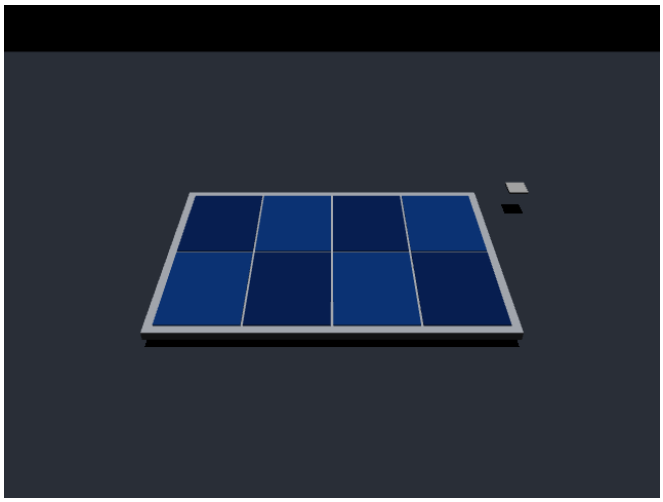
Capture (s)	V / I / reference	Sample delivery (s)	Returned observation
1.900	1.800 V / 0.558140 A / 900 W/m2	1.920	#3 at 2.012 s
2.000	1.800 V / 0.304132 A / 500 W/m2	2.020	#4 at 2.512 s
2.100	1.800 V / 0.304088 A / 500 W/m2	2.120	#4 at 2.512 s

Re-extracted from retained public observations, not reconstructed from truth [A2].

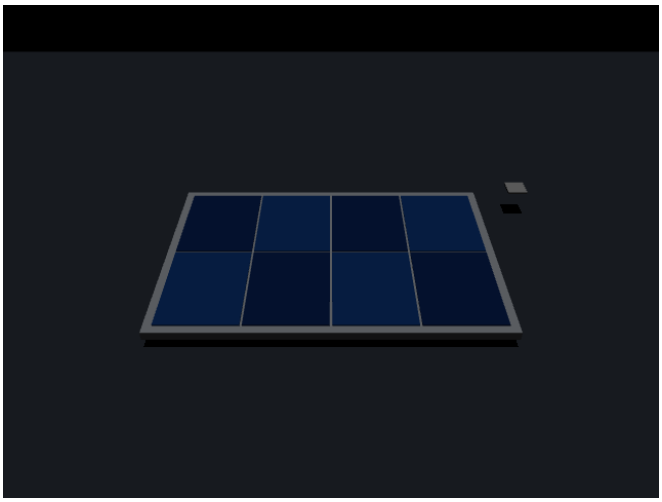
The 2.000 s measurement is delivered to the sensor buffer at 2.020 s, but in this fixture it reaches the caller in observation #4 at 2.512 s. A policy invoked only at those decision boundaries could not claim to have responded at 2.020 s. Detection delay must specify whether it means capture, sensor delivery, observation return or actual decision time. [A2]



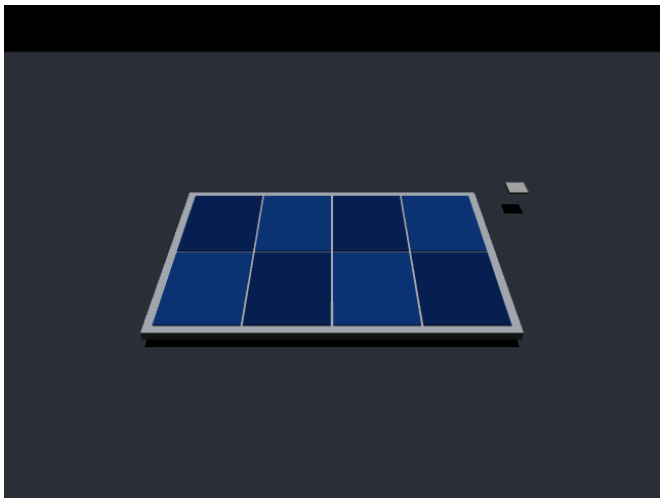
Treatment before event: capture 0.226 s; delivery 0.332 s.



Treatment pre-event capture 1.906 s; delivered after event at 2.012 s.



Treatment after event: capture 2.406 s; delivery 2.512 s.



Unchanged-source control at the same 2.406 s capture.

Figure 5. Original 640 x 480 MuJoCo RGB frames. The delayed pre-event frame retains its original lighting even though it arrives after the change. Only a new treatment capture becomes darker; the control stays unchanged. No image is generated or altered for this report.

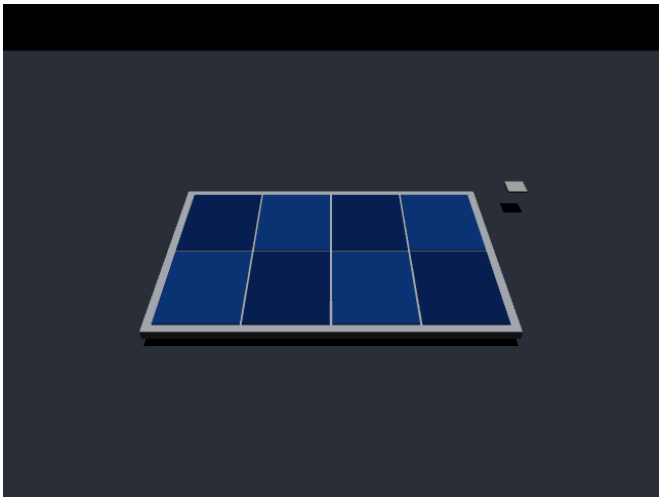
Accepted source-response desktop images and visual/integrity review [R18-R20].

The image captured at 1.906 s arrives at 2.012 s, after the source changed, yet still shows the earlier lighting. The image captured at 2.406 s shows the darker state. Looking only at arrival time would make the image appear to contradict the new sensor values. Correct reasoning must preserve when evidence was acquired. Brightness is qualitative renderer evidence, not a calibrated irradiance measurement. [R19, A2]

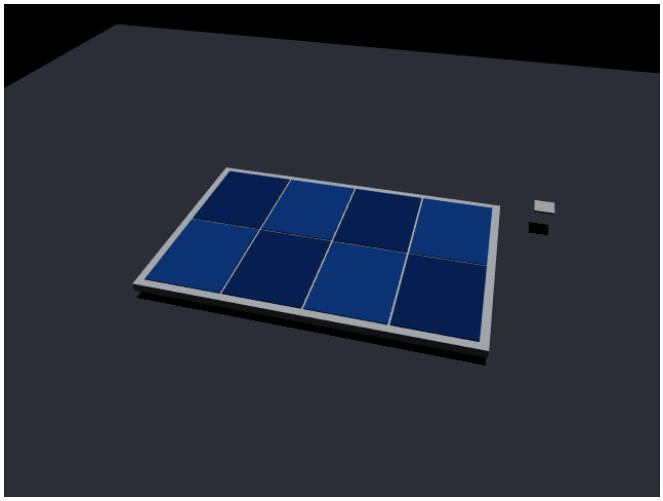
What was identified here?
The evaluator and independent calculations verify the imposed change and its consequences. No agent identified it in this retained source fixture. The reference sensor makes the change observable, but observability is not a recorded inference. A future agent test must retain the agent's own conclusion before exposing the private answer.

6. AGENT test: tools, skills, actions and WORLD responses

The accepted B1 seed-11 episode demonstrates a limited AGENT loop in a nominal WORLD. Unlike the source-response fixture, B1 actually receives public observations and issues decisions. Its sequence of investigations is fixed; its operating-voltage choice depends on measured power. The source event in Section 5 does not occur in this episode. [R11-R12, A1-A2]



M1: actual home camera capture



M1: actual side camera capture

Figure 6. Original 640 × 480 camera images from the accepted M1 pilot. The side capture occurred at 3.466 s with actual rail position 0.1199964 m and yaw 0.2606024 rad. The images differ at 40,256 pixel locations, consistent with changed viewpoint.

Accepted public frames [R11]; independent motion/image audit [R2, R4].

Table 12. Six AGENT decisions, with their tools, skills and returned evidence.

Issue (s)	AGENT skill / procedure	Tool and chosen action	WORLD / SENSORS result used
0.020	Check declared availability; acquire a view.	Camera tool: capture home RGB.	The image is returned at 0.332 s. B1 checks availability, not image semantics.
0.332	Acquire another reachable view and confirm motion.	Motion/camera tool: move 0.12 m, yaw toward panel, capture.	Finite WORLD motion; settled encoder evidence and side RGB returned at 3.572 s.
3.572	Measure comparable probe points.	Load tool: probe 0.8/1.8/3.0 V with fixed dwell.	Actual delay/slew, measured V/I and restored prior voltage; observation returned at 7.020 s.
7.020	Pair timestamps and rank measured V x I.	Load tool: request selected 1.8 V.	Fresh measured voltage confirmation returned at 7.620 s.
7.620	Seek evidence newer than the setting confirmation.	Wait tool: collect another 0.5 s of WORLD evolution.	New V/I captured at 8.300 s and returned at 8.320 s.
8.320	Check the new pair and keep the diagnosis unresolved.	Finish tool: retain 1.8 V through verification tail.	WORLD continues until 10.520 s; final sensor confirmation is recorded.

Actual decision times, public feedback and evidence references [A2, A5]. 'Skill' denotes a programmed procedure, not learned competence.

Table 13. Actual delivered probe results from the retained recording.

V (V)	I (A)	P (W)	Capture / delivery (s)	T at capture (°C)
0.800	0.748196	0.598557	4.600 / 4.620	31.981
1.800	0.553316	0.995969	5.500 / 5.520	33.275
3.000	0.317347	0.952041	6.400 / 6.420	34.543

All three are returned in the containing observation at 7.020 s. Temperature is evaluator truth shown for analysis, not an extra simultaneous policy action. [R12]

The best measured candidate is 1.8 V: 0.995969 W versus 0.598557 W at 0.8 V and 0.952041 W at 3.0 V. All three dwell records reach B1 together in observation #3 at 7.020 s. Probe restoration is checked separately and is not counted as a candidate. This demonstrates use of measured outcomes rather than a true-power oracle. [A2, A5]

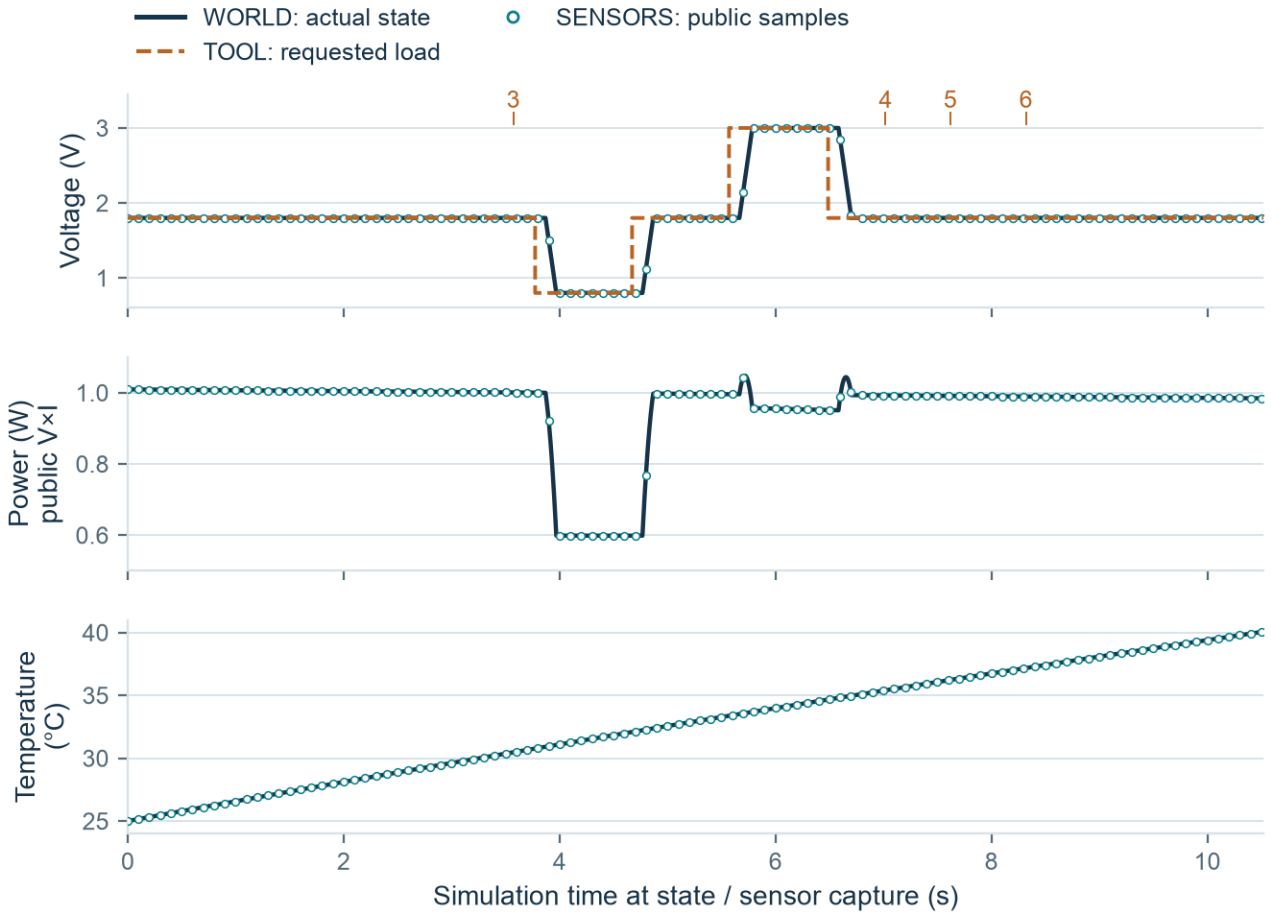
After applying the choice, B1 requests another wait. The newer confirming pair is captured at 8.300 s and returned at 8.320 s: 1.800 V and 0.549635 A. That observation is explicitly cited by Finish. A command saying 'set 1.8 V' would not be sufficient evidence. [A2]

B1 AGENT commands and WORLD responses: nominal episode

AGENT tools / skills: fixed scan → measured power choice → fresh verification

Ordered decisions below are evenly spaced; actual issue times are labelled.

1. Capture	2. Move/view	3. Probe	4. Choose/set	5. Verify wait	6. Finish
0.020 s	0.332 s	3.572 s	7.020 s	7.620 s	8.320 s



Tool request and world response differ during delay/slew. Samples are placed at capture.

B1 checks newer V/I; no source-event detection, image interpretation or predictive model.

Figure 7. AGENT command/skill stages above the WORLD's actual physical response; public SENSOR measurements are separate markers. A requested voltage can differ from the applied voltage during delay and slew. The AGENT's 1.8 V choice uses delivered probe evidence, and its final check uses a later V/I pair.

Read-only replot of the accepted B1 episode [R11-R12, A2, A7]; no new policy run.

The causal link demonstrated by B1

The AGENT requests a load change; the WORLD applies it through finite dynamics; SENSORS return V/I; the AGENT compares measured powers, chooses an operating point and checks fresh evidence. This is procedural sensing and decision making. It is not a source-change explanation, learned world model, semantic image skill or experience-memory retrieval.

7. INTERFACE test: recorded information linked to physical actions

This fixture tests an additional AGENT-facing tool interface. A human/internet request has a question, information gap and intended decision; the returned record has attribution and availability. In the accepted test, an authored DRIVER fills the agent-facing role, the ADAPTER returns simulated replies, and the WORLD executes a bounded voltage probe through the action gate. No actual human, internet service or adaptive agent chooses these exchanges. [R21-R24]

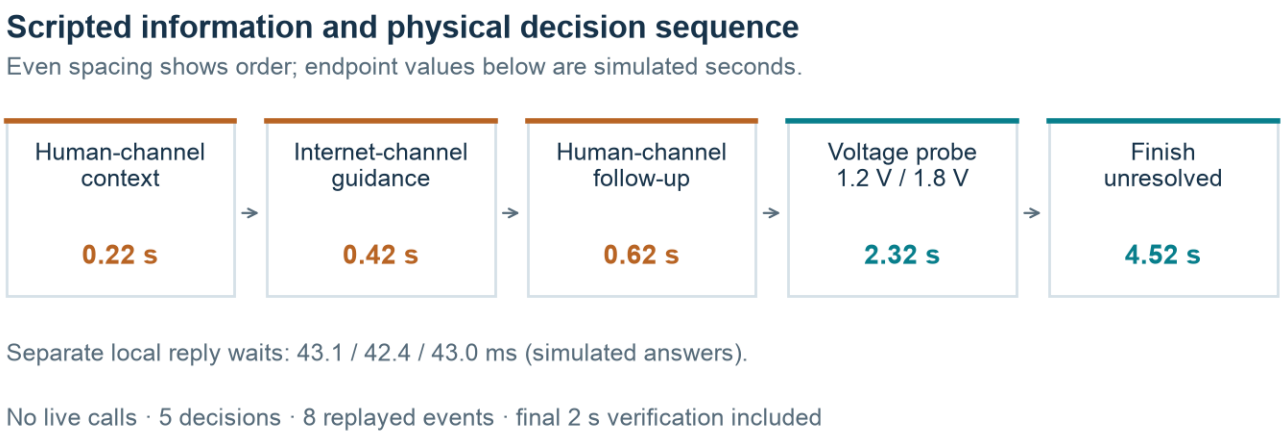


Figure 8. Sequence of the retained scripted information fixture. Endpoint labels use simulated time; displayed waiting is actual local computation/fixture delay and is not added to simulated time. The sequence is authored, not evidence of autonomous information choice.

Public interaction transcript and fixture summary [R22]; read-only diagram [R26].

Table 14. Questions, evidence and the later decision.

Exchange	Authored fixture content	Interpretation
Human channel: context	Hypothetical report that the surface appeared briefly less bright, without an instrument or controlled viewpoint.	An uncertain report, not an evaluator label or verified cause.
Internet channel: guidance	Use allowed voltage points and fresh matched V/I confirmation after settling.	Generic test guidance; no retrieval actually occurred.
Human channel: follow-up	Clarifies that the earlier brightness impression was qualitative only.	The fixture retains uncertainty rather than declaring the cause known.
Separate physical action	Probe 1.2 and 1.8 V, verify fresh measurements and restoration; Finish unresolved.	Execution success is distinguished from diagnostic success.

All three replies have simulated provenance and an authored source record [R22].

The retained fixture makes three authored requests, runs a two-point 1.2/1.8 V probe, then finishes unresolved. It runs under constant nominal physical conditions. The driver receives the replies but does not choose new questions or actions by reasoning about their content. Editing the prose after a run would not turn this into adaptive interaction. [R22, A1]

Table 15. What the interface now enforces.

Property	Implemented behaviour	Scientific implication
Provenance	Only recorded/simulated replies; attributed available answers and explicit failures.	No accepted live human, web or model-provider exchange.
Budgets	Three shared external requests; each follow-up counts; total decision cap also applies.	Information is a constrained resource, not a free oracle.
Time	Each request advances 0.2 simulated seconds before pausing at a settled boundary. Waiting freezes physics and event delivery.	Record actual wall waiting separately. This is offline experimentation, not real-time energy saving.
Control	Response text is not executable and is not directly parsed into a physical command.	A separate validated decision must pass the existing physical gate.
Replay	Eight events reproduce public records and metrics; 4.52 simulated seconds in the retained fixture.	Deterministic replay verifies interface accounting, not correctness of external knowledge.

The missing reasoning test
Supply two relevant replies under the same public physical history and check whether the declared agent appropriately changes its next question or physical test. Supply an irrelevant or unsupported claim and check that it does not force an unjustified action. Retain the pre-action prediction and later measurement. This is planned agent evaluation, not a result of the present fixture.

In a future live system, a human or web source would be outside the agent, just as the physical plant is outside it. Their replies would enter as attributed evidence, not as executable commands or guaranteed truth. The internal agent process would decide whether the reply changes its belief, next question or action. Current information_refs link records to decisions but do not implement that reasoning.

8. Tests organized by the interaction loop

The retained test guide organizes existing tests by the claim they protect. Its exact test selectors and machine-readable map are unchanged. V3 carries forward the focused v2 execution and accepted desktop render evidence; no test suite is rerun for these explanatory edits. World-mechanism checks, policy-decision checks and future memory tests remain distinct. [A3]

> **Retained v2 focused test result**
> 54 passed in 9.73 s; zero failures, errors or skips in the v2 report task. The 41 selected existing functions expanded to 54 cases (26 integration, 23 unit, 5 security). All 61 frozen inputs were unchanged. No render test ran in that subset. These results are reused as evidence, not reported as new v3 execution. [A3]

Table 16. What each test stage establishes and what remains unproven.

Stage	Current executable coverage	Limit of the claim
1. Hidden world	Source schedule boundaries, old-interval/new-endpoint ordering, independent source-response and thermal reference checks.	Implemented model consistency; no physical-panel agreement.
2. Sensing	Capture versus delivery, held snapshots, channel values, validity and stale/missing samples; retained actual camera checks.	Accessible evidence is correct; no inference follows automatically.
3. Public boundary	Serialized policy inputs; private fields rejected/absent; source schedule kept out of public logs.	Trusted-code boundary, not a malicious-code sandbox.
4. Analysis/decision	B1 chooses maximum measured probe power and rejects inadequate confirmation; no arbitrary near-target sample substitution.	Operating-point ranking and procedural stopping; no causal explanation or channel-selection algorithm.
5. Action execution	Delay/slew, encoder settling, dropped/delayed commands, restoration and timeout cleanup.	Requested, applied and observed states remain distinct.
6. Verification	Fresh matched samples, evidence references, no premature completion, Finish/safe-stop behaviour.	Verifies the chosen action outcome; diagnosis may remain unresolved.
7. Information	Contract/budget/retry/pause rules, attribution and action-gate boundary, fixture replay.	The scripted fixture cannot prove response-dependent reasoning or source usefulness.
8. Whole loop	Separate component and pilot evidence above.	No accepted source-change-to-adaptive-decision episode or matched policy-utility study yet.

Historical counts are checkpoints, not independent experiments: Step 1 has 599 passing cases including 34 render cases; Step 2 has 708 non-render cases. They overlap and must not be added. The v2 focused rerun is another subset, not an increase in the accepted milestone count. Exact current node IDs, outcome counts and source hashes are recorded in the test map and JUnit evidence. [R18, R21, A3]

What would fail if the implementation cheated?

Table 17. Adversarial interpretations that the tests must reject.

Bad shortcut	Required observation
Reveal the future event or true fault to the policy	Only public task/history cross the policy boundary; logs contain no private schedule.
Read current state when delivering an old packet	The old packet retains its original value/image and capture time across an event.
Treat command acceptance as success	Without fresh measured target/pose evidence, the action must not be reported as verified.
Choose voltage using private optimum	B1 selection must match the delivered probe measurements, with correct evidence references.
Label scripted dialogue as autonomous reasoning	Require a response-dependent test before claiming adaptive interaction; fixture replay alone is insufficient.

9. The next integrated experiment to demonstrate identification

The next test should connect the existing pieces in one bounded episode. It need not introduce cleaning, reconfiguration, continual learning or new physics. Declare the decision agent, its public prior knowledge and its action tools; then require an auditable change-to-decision trace. This section is an experimental specification, not an executed result. [R17]

Table 18. Minimum trace to retain for every episode.

Record	Agent side	Evaluator side
Before change	Initial public history, known capabilities, budgets and operating goal.	Freeze initial state, private event schedule, noise/fault conditions and acceptable outcomes.
Change / observe	Receive timestamped public evidence; identify a discrepancy or say why none is resolvable.	Record actual state evolution and first time evidence could be returned; do not disclose the cause.
Interpret	State supporting observations, alternative causes and missing information.	Score attribution separately from safe operational usefulness.
Choose interaction	Select a local observation, physical probe, question, lookup or justified abstention.	Check equal availability and budget; do not reward unnecessary requests.
Predict / act	Commit expected observable effect and its time window before the measurement arrives.	Record accepted command, actual application and any failure.
Verify / revise	Compare fresh observations with prediction; retain or revise the explanation and operation.	Score prediction error, decision validity, costs and unresolved outcome against predeclared criteria.

A small test family, not an expanded fault taxonomy

Table 19. Proposed agent tests; none is claimed as passed here.

ID	Condition	Behaviour to test
P01	Accepted uniform source change and unchanged control; all existing sensors available.	Recognize the change using evidence available by decision time; measure detection delay and false alarms.
P02	Source change versus existing view-only obstruction.	Distinguish a blocked view from electrically effective shading; do not infer a power loss from appearance alone.
P03	Same public physical history, with relevant, irrelevant, unavailable or contradictory replies.	Show appropriate response-dependent follow-up or tests, with justified invariance to irrelevant claims.
P04	Local evidence sufficient versus a genuine information gap.	Choose sensors/probe/human/web or justified abstention; do not force outside help when measurements suffice.
P05	Advertised capability limit or measured failure during an action.	Update the effective capability assessment and replan; never confuse ineffective actuation with a panel fault.

ID	Condition	Behaviour to test
P06	Permitted physical probe with a prediction committed beforehand.	Compare fresh measurements with expected direction, magnitude and timing; score uncertainty or abstention.
P07	Matched fixed-sequence and declared-agent trials.	Compare decision validity, verification and resource costs under identical channels, limits and time treatment.
P08	Separate specified physical counterpart and held-out intervention data.	Measure absolute/change residuals after calibration and freezing; no current hardware agreement claim.

P03-P04 need a concrete decision whose acceptable answer truly depends on the supplied context. Define this before running the study. A manual may suggest a test without proving this panel's condition; a human may report recent work without knowing the electrical cause. If no such decision exists in the chosen simple scenario, report that outside information adds no demonstrated value instead of manufacturing an information gap. IDs P01-P08 match the accompanying test guide. [A3]

A minimal proposed world/self-model update example

Before a probe, the agent could record: 'Reference irradiance fell while applied voltage remained near 1.8 V; a source change is consistent with lower current. I do not yet know the external cause. The load action is advertised, but I will require measured voltage confirmation. I will probe the allowed points, compare delivered powers, then verify the selected point.' The first two sentences concern world belief; the capability and confirmation rule concern the self model. This is illustrative proposed reasoning, not a quotation from a current agent run.

> **Completion criterion**
> A convincing result is a retained agent-authored trace in which evidence available at that time supports a decision, the action is actually applied or its failure is recognized, and fresh measurements test the predicted outcome. Correctly concluding 'unresolved with these tools' is a valid outcome when the evidence cannot identify the cause.

Memory need not add a new training programme: the first declared agent can retrieve permitted experience while keeping its decision model fixed. Freeze the accessible memory collection and test set separately. A replayed old action is not a new adaptive choice; require evidence of applicability checking and verification in the current WORLD before claiming useful experience reuse.

10. Evaluate the agent and validate the physical world separately

There are two distinct research questions: whether the digital environment reproduces the relevant physical consequences, and whether the agent makes useful decisions from restricted evidence. Strong software tests support the first question only within the implemented equations; an intelligent-looking explanation is not a measurement of physical fidelity. [R0]

Table 20. A compact evaluation dashboard for the integrated study.

Dimension	Measure	Avoid
Identification	Correct explanation or equivalence class; appropriate unresolved result; time from first available evidence to decision.	Scoring a label supplied by the evaluator; conflating capture time and policy response time.
Prediction	Precommitted effect direction/magnitude/timing, residual and uncertainty coverage where declared.	Post-hoc rationales or predictions copied from private truth.
Action/verification	False completion, restoration success, fresh V/I/pose evidence and supported final decision.	Treating requested targets or acknowledgements as measurements.
Information use	Useful requests, source relevance, response sensitivity, contradictions and unnecessary queries.	Rewarding the number of questions or channels used.
Benefit/cost	PV energy over a common horizon; frames, probe points, travel, requests and failures; wall time separately.	Adding paused human waiting to simulated generation or comparing unequal episode horizons.
Robustness	Per-case outcome, repeated independent conditions, uncertainty and failure cases.	Treating thousands of time samples as independent episodes or a few deterministic seeds as general performance.

Match a digital intervention to a real panel intervention

Choose a physical counterpart first: panel topology, electrical load, illumination range, temperature and sensing equipment. Calibrate on designated data, freeze the model and timing rules, then validate held-out illumination/load changes. Record actual V/I, irradiance and temperature together. Compare both absolute signals and changes: a model can predict a difference while retaining a large absolute bias. Preserve delayed responses and acquisition windows. [S7, S9]

Table 21. Matched physical validation examples.

Change	Digital comparison	Independent physical evidence needed
Source illumination step	Current/power amplitude, unchanged voltage, continuous temperature and subsequent thermal slope.	Measured illumination at the panel, terminal V/I and temperature time series; stable or characterized spatial/spectral conditions.
Load voltage probe	Actual applied voltage, settling, V/I/P at each dwell and restoration.	Load timing/settling and calibrated electrical acquisition; account for changing temperature/irradiance during the scan.

Change	Digital comparison	Independent physical evidence needed
Camera move/view obstruction	Measured pose, reprojection, visibility, capture/delivery and stale-image handling.	Independent pose/geometry and camera calibration. View obstruction alone must not be interpreted as reduced PV illumination.
Failure to apply	Requested versus measured action and bounded cleanup.	Independent hardware command/measurement records under a supervised permitted failure test.

Report bias, RMSE, worst error, change residuals, transient delay and integrated energy with units. Use an uncertainty budget that includes shared calibration/covariance and model discrepancy. Freeze acceptance margins from the intended decision; software equality tolerances are not physical accuracy claims. The detailed calibration, hold-out and uncertainty protocol remains in technical report v1. No hardware validation or measured dataset analysis has been added by this reorganization. [R0, S10-S11]

11. Development status and the next bounded prototype

Table 22. Current evidence versus the next claim.

Status	Evidence / work	Permitted conclusion
Implemented	Private physical simulator and public observation/action contracts; accepted M0-M2, source scenario and information interface.	An inspectable simulation and interaction foundation.
Demonstrated separately	Hidden source-change response; B1 measured voltage choice; scripted information-to-probe/replay fixture.	Individual causal and interface links have retained evidence.
New in v3	Four comments addressed: separated WORLD/AGENT/EVALUATOR roles, memory without training, and tool/skill/action labels on retained traces.	A clearer architecture and evidence explanation. No new runtime or memory capability is claimed.
Next Step 3	Declare the agent; demonstrate response-dependent decisions and actual human/web channels under the approved plan.	A bounded adaptive interaction demonstration only after its trace and outcomes are reviewed.
Then Step 4	Frozen comparator, matched evidence/budgets and predeclared outcome rubric.	Measured usefulness within the tested cases; no automatic general superiority.
Separate work	Real-panel calibration and held-out intervention validation.	A bounded physical fidelity claim only after independent measurements.

The source-change experiment should remain the simplest local-evidence case. Develop the response-dependent context case only where it changes a legitimate decision. Add explicit agent world-belief and capability records when they help test prediction or replanning; do not require a complex learned architecture for the first prototype. This preserves the high-level physical-experiment versus external-information question while keeping the implementation small. [R17]

Reproducibility and reading guide

The original report, v1 and annotated v2 remain unchanged. This version preserves a snapshot and inventory of all four comments under runs/comment-report-v3-20261019/inputs/ and analysis/. Editable content, plotting scripts, comment responses and frozen PDF inputs are retained alongside them. The v2 test guide/results remain authoritative for the 54-case subset. No runtime code, test selection, acceptance record or agent capability is changed. [A3, C1]

All camera images are retained original render evidence. The new architecture diagrams are explanatory drawings, not sensor images. In every story, evaluator truth, public measurements, agent decisions and future design are labelled separately. Documentation of a possible world/self model is not evidence that the present agent has learned or updated one.

12. Response to your four comments in v2

All four text comments were extracted from the annotated v2 PDF and checked against their highlighted pages. This table records how each is addressed. The annotations and their source hash are preserved; no comment is treated as a new experimental result. [C1]

Table 23. Comment-to-revision map.

Comment / v2 page	Requested clarification	Revision in v3
C01 / p. 8	Separate the non-agent world from internal agent models, including replacement by the real world.	Sections 1, 2 and 4 distinguish WORLD/plant, INTERFACE, AGENT and EVALUATOR. Figure 1 shows the boundaries. Hardware substitution is an architectural intention with explicit adapter/timing validation, not an implemented capability.
C02 / p. 9	Clarify whether records of past inputs, choices and outcomes can guide future actions without training a new model.	Section 4 answers yes as a proposed memory approach: record, retrieve, check relevance, reason, act and verify. It separates implemented recording/history from unimplemented retrieval/reuse, and memory updates from weight training. It specifies useful record fields and tests.
C03 / p. 10	Label the world, agent and interaction in the source-change example.	Section 5 labels the EVALUATOR-imposed change, WORLD dynamics, SENSOR evidence and absence of an AGENT decision in that fixture. The new source plot and ownership table keep the unexecuted agent step explicit.
C04 / p. 15	Label world and agent and highlight the agent's tools, skills and actions.	Sections 3 and 6 distinguish tools, programmed procedures, specific actions and physical outcomes. The six-decision table and new B1 trace align AGENT commands with WORLD response and public SENSOR measurements.

Scope of this revision
Conceptual explanations, diagrams and evidence labels have changed. Existing data, agent code and tests have not. The WORLD/AGENT architecture, proposed memory mechanism and future physical adapter must still be demonstrated through the planned tests before stronger claims are made.

References and evidence sources

Repository paths identify retained local evidence. External references provide methodological context; they do not substitute for measurements of the intended hardware.

[R0] Technical methods report, version 1. Preserved detailed equations, numerical validation and physical validation protocol from the earlier technical report.

output/pdf/technical_report_v1.pdf

[R2] M1 accepted rendered loop and replay. Accepted 408-case suite, three retained B1 pilots and replay evidence.

docs/evidence/M1-desktop-acceptance-20261017.md

[R4] Historical M0-M2 development and test evidence map. 13-family, 93-function map at the 18 October checkpoint; not the current 708-case inventory. Human-readable companion: development-test-map.txt.

runs/development-explanation-20261018/analysis/development-test-map.json

[R7] M1 runtime conventions. Shared clock, optical/electrical/thermal conventions, actions and sensing.

docs/decisions/0008-m1-runtime-conventions.md

[R8] M2 integrity conventions. Noise/transport/fault semantics and completion rules.

docs/decisions/0011-m2-integrity-conventions.md

[R9] Accepted M2 rendered report provenance. Original geometry plot, render evidence and report source snapshots.

runs/m2-desktop-report-20261018/provenance.json

[R11] Accepted M1 seed-11 recording. Actions/feedback/frames are in the same public directory. Private truth and metrics are in private_eval/m1-desktop-20261017T192106Z-8055-s11/.

runs/m1-desktop-20261017T192106Z-8055-s11/public/observations.jsonl

[R12] Read-only retained-run analysis. Source hashes, sample residuals, probe values and metrics; companion analysis script, CSV and chart-contracts.json.

runs/development-explanation-20261018/analysis/retained-run-analysis.json

[R17] Approved interactive PV prototype plan. One asset; choice between physical experiments and outside information; completed Steps 1-2 and planned Steps 3-4.

docs/planning/interactive-pv-prototype-v1.md

[R18] Controlled source-response acceptance. 599 full tests, actual camera evidence and bounded interpretation; historical next-step wording is superseded by R21.

docs/evidence/source-response-desktop-acceptance-20261019.md

[R19] Retained paired source-response fixture. Control/treatment public frames; private_eval companion contains 4,001-row truth and independent-reference arrays per case.

runs/source-desktop-20261019T055548Z-93694-source-response/report.json

[R20] Independent source-response desktop review. PASS_STEP1; independently reimplemented reference and checked retained numerical/visual evidence.

output/review/SOURCE_RESPONSE_DESKTOP_REVIEW_20261019.txt

[R21] Information-interface acceptance. PASS_STEP2_INTERFACE, 708 non-render tests, frozen evidence and explicit offline-only scope.

docs/evidence/information-interface-acceptance-20261019.md

[R22] Retained information fixture and exact replay. Public interaction.json and replay.json beside the summary; private companion retains configuration, inputs and metrics.

runs/interaction-step2-final-20261019T061833Z-fixture/summary.json

[R23] Information interface scope and frozen limits. Companion configs/interaction_protocol.json and docs/interaction-interface.md define API behavior and limitations.

docs/decisions/0013-information-interface-scope.md

[R24] Independent information-interface review. Four findings closed; separate review-input JSON identifies 68 checked files.

output/review/INTERACTION_INTERFACE_REVIEW_20261019.txt

[R26] Version 1 read-only analysis and report source. New figures, chart definitions and input hashes; author_v1.py/content.json provide editable report content, with build inputs and QA under report/.

runs/technical-report-v1-20261019/analysis/new-evidence-analysis.json

[A1] Current agent and model capability audit. Source-backed implemented/proposed distinctions; no runtime changes.

runs/interaction-report-v2-20261019/analysis/current-agent-inventory.json

[A2] Read-only causal-loop record analysis. Source sample/buffer/return timing and six B1 decisions; original input hashes and analysis script retained.

runs/interaction-report-v2-20261019/analysis/loop-record-analysis.json

[A3] Interaction-loop test guide and map. Companion interaction-test-map.json and runs/interaction-report-v2-20261019/analysis/focused-test-results.json retain exact selections and focused JUnit evidence.

docs/testing/interaction-test-guide.md

[A4] Public task, samples and decision contracts. Projection in src/pv_sim/sensors.py; policy input validation in src/pv_sim/policy_runner.py.

src/pv_sim/contracts.py

[A5] Implemented scan policy and public baseline helpers. FixedPolicy and protocol.py in the same directory; no causal inference, model training or information-channel policy.

src/pv_sim/policies/scan.py

[S7] Photovoltaic Calibrations at the National Renewable Energy Laboratory and Uncertainty Analysis Following the ISO 17025 Guidelines. NREL/TP-5J00-66873, September 2016 Rechecked 19 October 2026. Reference-condition IV characterization accounts for spectrum, temperature, irradiance, variable-load measurements, reference devices and sample-specific uncertainty. Full report accessed through official DOE OSTI; previous NLR PDF link timed out.

https://www.osti.gov/servlets/purl/1328360

[S9] Photovoltaic Module Energy Yield Measurements: Existing Approaches and Best Practice. Report IEA-PVPS T13-11:2018, May 2018. Rechecked 19 October 2026. Four-wire DC measurement, appropriate load response and sweep settings, acquisition synchronization and uncertainty reporting are important measurement-design considerations.

https://iea-pvps.org/wp-content/uploads/2020/01/Photovoltaic_Module_Energy_Yield_Measurements_Existing_Approaches_and_Best_Practice_by_Task_13.pdf

[S10] NIST Technical Note 1297: 5. Combined Standard Uncertainty. Rechecked 19 October 2026. Combination of standard uncertainty components with covariances as appropriate.

https://www.nist.gov/pml/nist-technical-note-1297/nist-tn-1297-5-combined-standard-uncertainty

[S11] NIST Technical Note 1297: 7. Reporting Uncertainty. Rechecked 19 October 2026. Report uncertainty components and their evaluation, and the coverage factor for expanded uncertainty; probability interpretations need a justified basis.

https://www.nist.gov/pml/nist-technical-note-1297/nist-tn-1297-7-reporting-uncertainty

[A6] Recording, policy history and information transcript implementations. Also src/pv_sim/policy_runner.py, policies/protocol.py and interaction.py. Durable records and per-episode history exist; these are not semantic cross-episode memory retrieval.

src/pv_sim/recorder.py

[A7] V3 role-labelled scientific trace plots. Retained source/B1 inputs, numeric checks and figure hashes; no simulator run.

runs/comment-report-v3-20261019/analysis/role-evidence-figure-provenance.json

[A8] V3 world-agent and memory architecture figures. New explanatory diagrams with implemented and proposed capabilities distinguished.

runs/comment-report-v3-20261019/analysis/concept-figure-provenance.json

[C1] Wei's four annotations and their resolution. Exact comments from annotated v2 pages 8, 9, 10 and 15. Preserved PDF snapshot in inputs/; comment-response.json records the revisions.

runs/comment-report-v3-20261019/analysis/annotation-inventory.json